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KOTA (RAJASTHAN)

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# JEE (MAIN+ADVANCED)

# Mathematics

## ENTHUSIAST COURSE



**STUDY MATERIAL**

## Differential Equation & Area Under the Curve

English Medium

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# DIFFERENTIAL EQUATION

## 1. DIFFERENTIAL EQUATION :

An equation that involves independent and dependent variables and the derivatives of the dependent variables is called a **differential equation**.

A differential equation is said to be **ordinary**, if the differential coefficients have reference to a single independent variable only e.g. $\frac{d^2y}{dx^2}-\frac{2dy}{dx}+\cos x=0$ and it is said to be **partial** if there are two or more independent variables. e.g. $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=0$ is a partial differential equation. We are concerned with ordinary differential equations only.

## 2. ORDER OF DIFFERENTIAL EQUATION :

The order of a differential equation is the order of the highest differential coefficient occurring in it.

## 3. DEGREE OF DIFFERENTIAL EQUATION :

The highest exponent of the highest order differential coefficient, when the differential equation is expressed as a polynomial in all the differential coefficient is called degree of the differential equation.

Thus the differential equation :

$$a_{0,0}+\sum_{i=1}^{n_1}a_{i,1}\left(\frac{d^m y}{dx^m}\right)^i+\sum_{i=1}^{n_2}a_{i,2}\left(\frac{d^{m-1}y}{dx^{m-1}}\right)^i+\ldots=0 \text{ is of order m \& degree } n_1.$$

where $a_{i,j}$ is some expression in x, y and $a_{n_1,1}\neq 0$

**Note :**

(i) The exponents of all the differential coefficient should be free from radicals and fraction.

(ii) The degree is always positive natural number.

(iii) The degree of differential equation may or may not exist.

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***Illustration 1 :*** Find the order and degree of the following differential equation :

$$\text{(i) } \sqrt{\frac{d^2y}{dx^2}}=\sqrt[3]{\frac{dy}{dx}+3} \quad \text{(ii) } \frac{d^2y}{dx^2}=\sin\left(\frac{dy}{dx}\right) \quad \text{(iii) } \frac{dy}{dx}=\sqrt{3x+5}$$

***Solution :***

(i) The given differential equation has the order 2. Since the given differential equation is not a polynomial in the differential coefficients, the degree of the equation is **not defined**.

(ii) The given differential equation has the order 2. Since the given differential equation is not a polynomial in the differential coefficients, the degree of the equation is **not defined**.

(iii) Its order is 1 and degree 1. **Ans.**

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***Illustration 2 :*** The order and degree of the differential equation $\left(\frac{d^2s}{dt^2}\right)^2 + 3\left(\frac{ds}{dt}\right)^3 + 4 = 0$ are -

(A) 2 , 2 (B) 2 , 3 (C) 3 , 2 (D) none of these

***Solution :*** Clearly order is 2 and degree is 2 (from the definition of order and degree of differential equations). **Ans. (A)**

**Do yourself - 1 :**

Find the order and degree of following differential equations

**1.** $[1 + (y')^2]^{1/2} = x^2 + y$ **2.** $(1 + y')^{1/2} = y''$ **3.** $y' = \sin y$

**4.** $\frac{d^2y}{dx^2} + \left[y + \left(\frac{dy}{dx}\right)^7\right]^{1/4} = 0$ **5.** $\frac{dy}{dx} + y^2 = \frac{1}{\frac{dy}{dx}}$ **6.** $e^{\frac{d^3y}{dx^3}} - x\frac{d^2y}{dx^2} + \frac{y}{2} = 0$

**7.** $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y + 3$ **8.** $\ell n\left(\frac{dy}{dx}\right) = ax + by + c$

**9.** $\left[\frac{d^2x}{dt^2}\right]^3 + \left[\frac{dx}{dt}\right]^4 - xt = 0$ **10.** $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$

## 4. FORMATION OF A DIFFERENTIAL EQUATION :

In order to obtain a differential equation whose solution is

$$f(x_1, y_1, c_1, c_2, c_3.....,c_n) = 0$$

where $c_1, c_2,.....c_n$ are 'n' arbitrary constants, we have to eliminate the 'n' constants for which we require (n + 1) equations.

A differential equation is obtained as follows :

(a) Differentiate the given equation w.r.t the independent variable (say x) as many times as the number of independent arbitrary constants in it.

(b) Eliminate the arbitrary constants.

(c) The eliminant is the required differential equation.

**Note :**

(i) A differential equation represents a family of curves all satisfying some common properties.

This can be considered as the geometrical interpretation of the differential equation.

(ii) For there being n differentiation, the resulting equation must contain a derivative of $n^{th}$ order

i.e. equal to number of independent arbitrary constant.

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***Illustration 3 :*** Find the differential equation of all parabolas whose axes is parallel to the x-axis and having latus rectum a.

***Solution :*** Equation of parabola whose axes is parallel to x-axis and having latus rectum 'a' is $(y-\beta)^2 = a(x-\alpha)$

Differentiating both sides, we get $2(y-\beta)\dfrac{dy}{dx} = a$

Again differentiating, we get

$$\Rightarrow 2(y-\beta)\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 0 \quad \Rightarrow \quad a\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^3 = 0. \qquad \textbf{Ans.}$$

***Illustration 4 :*** Find the differential equation whose solution represents the family : $c(y+c)^2 = x^3$

***Solution :*** $c(y+c)^2 = x^3$ ...(i)

Differentiating, we get, $c.\left[2(y+c)\right]\dfrac{dy}{dx} = 3x^2$

Writing the value of c from (i), we have

$$\frac{2x^3}{(y+c)^2}(y+c)\frac{dy}{dx} = 3x^2 \quad \Rightarrow \quad \frac{2x^3}{y+c}\frac{2x^3}{y+c} = 3x^2$$

$$\text{i.e. } \frac{2x}{y+c}\frac{dy}{dx} = 3 \Rightarrow \frac{2x}{3}\left[\frac{dy}{dx}\right] = y+c$$

$$\text{Hence } c = \frac{2x}{3}\left[\frac{dy}{dx}\right] - y$$

Substituting value of c in equation (i), we get $\left[\dfrac{2x}{3}\left(\dfrac{dy}{dx}\right) - y\right]\left[\dfrac{2x}{3}\dfrac{dy}{dx}\right]^2 = x^3,$

which is the required differential equation. **Ans.**

***Illustration 5 :*** Find the differential equation whose solution represents the family : $y = a\cos(\theta x) + b\sin(\theta x)$, where $\theta$ = fixed constant

***Solution :*** $y = a\cos(\theta x) + b\sin(\theta x)$, $\theta$ = fixed constant ....(i)

Differentiating, we get $\dfrac{dy}{dx} = -\theta a\sin(\theta x) + \theta b\cos(\theta x)$

Again differentiating, we get $\dfrac{d^2y}{dx^2} = -\theta^2 a\cos(\theta x) - \theta^2 b\sin(\theta x)$

using equation (i), we get $\dfrac{d^2y}{dx^2} = -\theta^2 y$ **Ans.**

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**Do yourself - 2**

**1.** Eliminate the arbitrary constants and obtain the differential equation satisfied by it.

**(i)** $y = 2x + ce^x$ **(ii)** $y = \left(\frac{a}{x^2}\right) + bx$ **(iii)** $y = ae^{2x} + be^{-2x} + c$

**2.** Form a differential equation for the family of curves represented by $ax^2 + by^2 = 1$, where a & b are arbitrary constants.

**3.** Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$ ; where g , f & c are arbitrary constants.

**4.** Obtain the differential equation associated with the primitive, $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$, where $c_1, c_2, c_3$ are arbitrary constants.

## 5. SOLUTION OF DIFFERENTIAL EQUATION :

The solution of the differential equation is a relation between the variables of the equation not containing the derivatives, but satisfying the given differential equation (i.e., from which the given differential equation can be derived).

Thus, the solution of $\frac{dy}{dx} = e^x$ could be obtained by simply integrating both sides, i.e., $y = e^x + c$ and that of, $\frac{dy}{dx} = px + q$ is $y = \frac{px^2}{2} + qx + c$, where c is arbitrary constant.

**(i)** **A general solution** or an integral of a differential equation is a relation between the variables (not involving the derivatives) which contains the same number of the arbitrary constants as the order of the differential equation.

For example, a general solution of the differential equation $\frac{d^2x}{dt^2} = -4x$ is $x = A\cos 2t + B\sin 2t$ where A and B are the arbitrary constants.

**(ii)** **Particular solution or particular integral** is that solution of the differential equation which is obtained from the general solution by assigning particular values to the arbitrary constant in the general solution.

For example, $x = 10\cos 2t + 5\sin 2t$ is a particular solution of differential equation $\frac{d^2x}{dt^2} = -4x$ .

**Note :**

**(i)** The general solution of a differential equation can be expressed in different (but equivalent) forms. For example

$$\log x - \log (y + 2) = k \quad \text{....(i)}$$

where k is an arbitrary constant is the general solution of the differential equation $xy' = y + 2$.

The solution given by equation (i) can also be re-written as

$$\log\left(\frac{x}{y+2}\right) = k \text{ or } \frac{x}{y+2} = e^k = c_1 \quad \text{...(ii)}$$

or $x = c_1(y + 2)$ ...(iii)

where $c_1 = e^k$ is another arbitrary constant. The solution (iii) can also be written as

$$y + 2 = c_2 x$$

where $c_2 = 1/c_1$ is another arbitrary constant.



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**(ii)** All differential equations that we come across have unique solutions or a family of solutions.

For example, the differential equation $\left|\frac{dy}{dx}\right| + |y| = 0$ has only the trivial solution, i.e. $y = 0$.

The differential equation $\left|\frac{dy}{dx}\right| + |y| + c = 0,\ c > 0$ has no solution.

## 6. ELEMENTARY TYPES OF FIRST ORDER & FIRST DEGREE DIFFERENTIAL EQUATIONS :

### (a) Separation of Variables :

Some differential equations can be solved by the method of separation of variables (or "variable separable").

This method is only possible, if we can express the differential equation in the form $A(x)dx + B(y)\,dy = 0$ where $A(x)$ is a function of 'x' only and $B(y)$ is a function of 'y' only.

A general solution of this is given by,

$$\int A(x)\,dx + \int B(y)dy = c$$

where 'c' is the arbitrary constant.

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***Illustration 6 :*** Solve the differential equation $xy\frac{dy}{dx} = \frac{1+y^2}{1+x^2}(1+x+x^2)$.

***Solution :*** Differential equation can be rewritten as

$$xy\frac{dy}{dx} = (1+y^2)\left(1+\frac{x}{1+x^2}\right) \Rightarrow \frac{y}{1+y^2}dy = \left(\frac{1}{x}+\frac{1}{1+x^2}\right)dx$$

Integrating, we get

$$\frac{1}{2}\ell n(1+y^2) = \ell n\,x + \tan^{-1}x + \ell n\,c \Rightarrow \sqrt{1+y^2} = cxe^{\tan^{-1}x}.$$ **Ans.**

***Illustration 7 :*** Solve the differential equation $(x^3 - y^2x^3)\frac{dy}{dx} + y^3 + x^2y^3 = 0$.

***Solution :*** The given equation $(x^3 - y^2x^3)\frac{dy}{dx} + y^3 + x^2y^3 = 0$

Case-I : $y = 0$ is one solution of differential equation

Case-II : If $y \neq 0$, then

$$\Rightarrow \frac{1-y^2}{y^3}dy + \frac{1+x^2}{x^3}dx = 0 \quad \Rightarrow \int\left(\frac{1}{y^3}-\frac{1}{y}\right)dy + \int\left(\frac{1}{x^3}+\frac{1}{x}\right)dx = 0$$

$$\Rightarrow \log\left(\frac{x}{y}\right) = \frac{1}{2}\left(\frac{1}{y^2}+\frac{1}{x^2}\right) + c$$

$$y = 0 \text{ or } \log\left(\frac{x}{y}\right) = \frac{1}{2}\left(\frac{1}{y^2}+\frac{1}{x^2}\right) + c$$ **Ans.**

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***Illustration 8 :*** Solve : $\frac{dy}{dx} = (x-3)\,(y+1)^{2/3}$

***Solution :*** Case-I : $y = -1$ is one solution of differential equation

Case-II : If $y \neq -1$, then

$$\frac{dy}{dx} = (x-3)(y+1)^{2/3}$$

$$\int \frac{dy}{(y+1)^{2/3}} = \int (x-3)dx$$

Integrate and solve for y : $3(y+1)^{1/3} = \frac{1}{2}(x-3)^2 + C$

$$(y+1)^{1/3} = \frac{1}{6}(x-3)^2 + C_0 \Rightarrow y+1 = \left(\frac{1}{6}(x-3)^2 + C_0\right)^3$$

$y = -1$ or $y = \left(\frac{1}{6}(x-3)^2 + C_0\right)^3 - 1$ **Ans.**

**Do yourself - 3 :**

Solve the following differential equations :

**1.** $\frac{2dy}{dx} = \frac{y(x+1)}{x}$

**2.** $\sqrt{1+4x^2}\,dy = y^3 x dx$

**3.** $(\tan y)\frac{dy}{dx} = \sin(x+y) + \sin(x-y)$

**4.** $\frac{\ell n(\sec x + \tan x)}{\cos x}dx = \frac{\ell n(\sec y + \tan y)}{\cos y}dy$

**5.** $(1-x^2)\,(1-y)\,dx = xy\,(1+y)\,dy$

**6.** $\frac{dy}{dx} + \frac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$

**7.** $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

**8.** $\frac{dy}{dx} = \frac{x(2\ell n x + 1)}{\sin y + y\cos y}$

**9.** **(a)** $\frac{dy}{dx} + \sin\frac{x+y}{2} = \sin\frac{x-y}{2}$ **(b)** $\sin x.\frac{dy}{dx} = y$ . *l*ny if $y = e$, when $x = \frac{\pi}{2}$

**10.** Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x\to\infty} y(x)$.

**11.** Let f(x) is a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t)dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{2}$. Find the value of $f\left(\sqrt{2}+1\right)$.

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**(i) Equation of the form :**

$\Rightarrow$ **$y' = f(ax + by + c), b \neq 0$**

To solve this, substitute $t = ax + by + c$. Then the equation reduces to separable type in the variable t and x which can be solved.

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***Illustration 9 :*** Solve $\frac{dy}{dx} = \cos(x + y) - \sin(x + y)$.

***Solution :*** $\frac{dy}{dx} = \cos(x + y) - \sin(x + y)$

Substituting, $x + y = t$, we get $\frac{dy}{dx} = \frac{dt}{dx} - 1$

Therefore $\frac{dt}{dx} - 1 = \cos t - \sin t$

$$\Rightarrow \int \frac{dt}{1 + \cos t - \sin t} = \int dx \Rightarrow \int \frac{\sec^2 \frac{t}{2} dt}{2\left(1 - \tan \frac{t}{2}\right)} = \int dx \Rightarrow -\ell n \left|1 - \tan \frac{x + y}{2}\right| = x + c.$$ **Ans.**

***Illustration 10 :*** Solve : $y' = (x + y + 1)^2$

***Solution :*** $y' = (x + y + 1)^2$ ....(i)

Let $t = x + y + 1$

$$\frac{dt}{dx} = 1 + \frac{dy}{dx}$$

Substituting in equation (i) we get

$$\frac{dt}{dx} = t^2 + 1 \Rightarrow \int \frac{dt}{1 + t^2} = \int dx \Rightarrow \tan^{-1} t = x + C \Rightarrow t = \tan(x + C)$$

$x + y + 1 = \tan(x + C) \Rightarrow y = \tan(x + C) - x - 1$ **Ans.**

**Do yourself - 4 :**

Solve the following differential equations :

**1.** $\frac{dy}{dx} = (y - 4x)^2$ **2.** $\tan^2(x + y)dx - dy = 0$

**3.** $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$ **4.** $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$ **5.** $\frac{dy}{dx} = (x + y)^2 - 1$

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**(ii) Equation of the form :** $\Rightarrow \dfrac{dy}{dx} = \dfrac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$

**Case I : If** $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$ **then**

Let $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \lambda$ then $a_1 = \lambda a_2$ .....(i) ; $b_1 = \lambda b_2$ .....(ii)

from (i) and (ii), differential equation becomes

$\dfrac{dy}{dx} = \dfrac{\lambda a_2x + \lambda b_2y + c_1}{a_2x + b_2y + c_2} \Rightarrow \dfrac{dy}{dx} = \dfrac{\lambda(a_2x + b_2y) + c_1}{a_2x + b_2y + c_2}$

or we can say, $\dfrac{dy}{dx} = f(a_2x + b_2y)$

which can be solved by substituting $t = a_2x + b_2y$

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***Illustration 11 :*** Solve : $(x + y)dx + (3x + 3y - 4)\,dy = 0$

***Solution :*** Let $t = x + y$

$\Rightarrow dy = dt - dx$

So we get, $tdx + (3t - 4)(dt - dx) = 0$

$2dx + \left(\dfrac{3t-4}{2-t}\right)dt = 0 \Rightarrow 2dx - 3dt + \dfrac{2}{2-t}dt = 0$

Integrating and replacing t by $x + y$, we get

$2x - 3t - 2[\ell n|(2 - t)|] = c_1$

$\Rightarrow 2x - 3(x + y) - 2[\ell n|(2 - x - y)|] = c_1$

$\Rightarrow x + 3y + 2\ell n|(2 - x - y)| = c$ **Ans.**

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**Case II :** **If $a_2 + b_1 = 0$, then** a simple cross multiplication and substituting d(xy) for xdy + ydx and integrating term by term, yield the results easily.

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***Illustration 12 :*** Solve $\dfrac{dy}{dx} = \dfrac{x - 2y + 1}{2x + 2y + 3}$

***Solution :*** $\dfrac{dy}{dx} = \dfrac{x - 2y + 1}{2x + 2y + 3}$

$\Rightarrow 2xdy + 2y\,dy + 3dy = xdx - 2y\,dx + dx$

$\Rightarrow (2y + 3)\,dy = (x + 1)\,dx - 2(xdy + ydx)$

On integrating, we get

$\Rightarrow \int(2y + 3)dy = \int(x + 1)dx - \int 2d(xy)$

Solving : $y^2 + 3y = \dfrac{x^2}{2} + x - 2xy + c$ **Ans.**



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**Do yourself - 5 :**

Solve the following differential equations :

**1.** $\frac{dy}{dx}=\frac{2x-y+2}{2y-4x+1}$ **2.** $\frac{dy}{dx}=\frac{3x-5y}{5x+y+3}$ **3.** $\frac{dy}{dx}=\frac{2x+3y-1}{4x+6y-5}$

**4.** $\frac{dy}{dx}=\frac{x-2y+5}{2x+y-1}$ **5.** $\frac{dy}{dx}=\frac{2x-y+3}{x+2y+4}$

**(iii) Equation of the form :**

$\Rightarrow$ **$yf(xy)dx + xg(xy)dy = 0$** ..... (i)

The substitution $xy = z$, reduces differential equation of this form to the form in which the variables are separable.

Let $xy = z$ ..... (ii)

$dy = \left[\frac{xdz - zdx}{x^2}\right]$ ..... (iii)

using equation (ii) & (iii), equation (i) becomes $\frac{z}{x}f(z)dx + xg(z)\left[\frac{xdz-zdx}{x^2}\right] = 0$

$\Rightarrow \frac{z}{x}f(z)dx + g(z)dz - \frac{z}{x}g(z)dx = 0 \quad \Rightarrow \quad \frac{z}{x}\{f(z)-g(z)\}dx + g(z)dz = 0$

$\Rightarrow \frac{1}{x}dx + \frac{g(z)dz}{z\{f(z)-g(z)\}} = 0$

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***Illustration 13 :*** Solve $y(xy + 1)dx + x(1 + xy + x^2y^2)dy = 0$

***Solution :*** Let $xy = v$

$y = \frac{v}{x} \Rightarrow dy = \frac{xdv - vdx}{x^2}$

Now, differential equation becomes $\Rightarrow \frac{v}{x}(v+1)dx + x(1+v+v^2)\left(\frac{xdv-vdx}{x^2}\right) = 0$

On solving, we get

$v^3dx - x(1 + v + v^2)\,dv = 0$

separating the variables & integrating we get

$\Rightarrow \int\frac{dx}{x} - \int\left(\frac{1}{v^3}+\frac{1}{v^2}+\frac{1}{v}\right)dv = 0 \quad \Rightarrow \quad \ell nx + \frac{1}{2v^2} + \frac{1}{v} - \ell nv = c$

$\Rightarrow 2v^2\ell n\left(\frac{v}{x}\right) - 2v - 1 = -2cv^2 \quad \Rightarrow \quad 2x^2y^2\ell ny - 2xy - 1 = Kx^2y^2$ where $K = -2c$

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**Do yourself - 6 :**

Solve the following differential equations :

**1.** $(y - xy^2)\,dx - (x + x^2y)\,dy = 0$ **2.** $y(1 + 2xy)\,dx + x(1 - xy)\,dy = 0$

**(iv) Transformation to polar-co-ordinates :**

Sometimes conversion of cartesian co-ordinates into polar coordinates helps us in separating the variables.

**(1)** $x = r\cos\theta,\ y = r\sin\theta$

then $x^2 + y^2 = r^2$

$xdx + ydy = rdr$

$xdy - ydx = r^2d\theta$

**(2)** $x = r\sec\theta,\ y = r\tan\theta$

then $x^2 - y^2 = r^2$

$xdx - ydy = rdr$

$xdy - ydx = r^2\sec\theta\,d\theta$

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***Illustration 14 :*** Solve : $\dfrac{x + y\dfrac{dy}{dx}}{y - x\dfrac{dy}{dx}} = x^2 + 2y^2 + \dfrac{y^4}{x^2}$

***Solution :*** The given equation can be reduced to

$$\frac{xdx + ydy}{ydx - xdy} = \frac{(x^2 + y^2)^2}{x^2}$$

Substituting $x = r\cos\theta$

$y = r\sin\theta$

we get, $\dfrac{rdr}{r^2d\theta} = \dfrac{-(r^2)^2}{r^2\cos^2\theta} \Rightarrow \int\dfrac{dr}{r^3} = -\int\sec^2\theta d\theta \Rightarrow -\dfrac{1}{2r^2} = -\tan\theta + c$

Substituting, $\dfrac{1}{2(x^2 + y^2)} = \dfrac{y}{x} + K$ **Ans.**

**Do yourself - 7 :**

Solve the following differential equations :

**1.** $x\,dx + y\,dy = xdy - y\,dx$ **2.** $y\,dx - x\,dy = xy\,dy - x^2\,dx$

**3.** $x\,dx + y\,dy = x\,(x\,dy - y\,dx)$ **4.** $\dfrac{xdx + ydy}{\sqrt{x^2 + y^2}} = \dfrac{xdy - ydx}{x^2}$

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**(b) Homogeneous equations :**

A function $f(x,y)$ is said to be a homogeneous function of degree n, if the substitution $x = \lambda x, y = \lambda y, \lambda > 0$ produces the equality

$$f(\lambda x, \lambda y) = \lambda^n f(x,y)$$

The degree of homogeneity 'n' can be any real number.

***Illustration 15 :*** Find the degree of homogeneity of function

(i) $f(x,y) = x^2 + y^2$ (ii) $f(x,y) = (x^{3/2} + y^{3/2})/(x+y)$ (iii) $f(x,y) = \sin\left(\frac{x}{y}\right)$

***Solution :*** (i) $f(\lambda x, \lambda y) = \lambda^2 x^2 + \lambda^2 y^2 = \lambda^2 (x^2 + y^2) = \lambda^2 f(x,y)$

degree of homogeneity $\to 2$

(ii) $f(\lambda x, \lambda y) = \dfrac{\lambda^{3/2} x^{3/2} + \lambda^{3/2} y^{3/2}}{\lambda x + \lambda y}$

$f(\lambda x, \lambda y) = \lambda^{1/2} f(x,y)$

degree of homogeneity $\to 1/2$

(iii) $f(\lambda x, \lambda y) = \sin\left(\dfrac{\lambda x}{\lambda y}\right) = \lambda^\circ \sin\left(\dfrac{x}{y}\right) = \lambda^\circ f(x,y)$

degree of homogeneity $\to 0$

***Illustration 16 :*** Determine whether or not each of the following functions is homogeneous.

(i) $f(x,y) = x^2 - xy$ (ii) $f(x,y) = \dfrac{xy}{x+y^2}$ (iii) $f(x,y) = \sin xy$

***Solution :*** (i) $f(\lambda x, \lambda y) = \lambda^2 x^2 - \lambda^2 xy = \lambda^2 (x^2 - xy) = \lambda^2 f(x,y)$ homogeneous.

(ii) $f(\lambda x, \lambda y) = \dfrac{\lambda^2 xy}{\lambda x + \lambda^2 y^2} \neq \lambda^n f(x,y)$ not homogeneous.

(iii) $f(\lambda x, \lambda y) = \sin(\lambda^2 xy) \neq \lambda^n f(x,y)$ not homogeneous.

**Do yourself - 8 :**

**1.** Find the degree of homogeneity of function $f(x,y) = x^3 \ell n\left[\sqrt{x+y} / \sqrt{x-y}\right]$

**2.** Find the degree of homogeneity of function $f(x,y) = ax^{2/3} + hx^{1/3} y^{1/3} + by^{2/3}$

**3.** Determine whether or not each of the following functions is homogeneous.

**(a)** $f(x,y) = \sqrt{x^2 + 2xy + 3y^2}$ **(b)** $f(x,y) = x + y\cos\dfrac{y}{x}$

**(c)** $f(x,y) = x\sin y + y\sin x$ **(d)** $f(x,y) = \cos(xy)$

**(i) Homogeneous first order differential equation**

A differential equation of the form $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$

where $f(x,y)$ and $g(x,y)$ are homogeneous functions of x,y and of the same degree, is said to be homogeneous. Such equations can be solved by substituting

$$y = vx,$$

so that the dependent variable y is changed to another variable v.

Since $f(x,y)$ and $g(x,y)$ are homogeneous functions of the same degree say, n, they can be written as

$$f(x,y) = x^n f_1\left(\frac{y}{x}\right) \quad \text{and} \quad g(x,y) = x^n g_1\left(\frac{y}{x}\right).$$

As $y = vx$, we have $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

The given differential equation, therefore, becomes

$$v + x\frac{dv}{dx} = \frac{f_1(v)}{g_1(v)} \Rightarrow \frac{g_1(v)dv}{f_1(v) - vg_1(v)} = \frac{dx}{x},$$

so that the variables v and x are now separable.

**Note :** Sometimes homogeneous equation can be solved by substituting $x = vy$ or by using polar coordinate substitution.

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***Illustration 17 :*** The solution of the differential equation $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x\cos y}$ is -

(A) $\sin^2 y = x\sin y + \frac{x^2}{2} + c$ (B) $\sin^2 y = x\sin y - \frac{x^2}{2} + c$

(C) $\sin^2 y = x + \sin y + \frac{x^2}{2} + c$ (D) $\sin^2 y = x - \sin y + \frac{x^2}{2} + c$

***Solution :*** Here, $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x\cos y}$

$\Rightarrow \cos y\frac{dy}{dx} = \frac{\sin y + x}{2\sin y - x}$, (put $\sin y = t$)

$\Rightarrow \frac{dt}{dx} = \frac{t + x}{2t - x}$ (put $t = vx$)

$$\frac{xdv}{dx} + v = \frac{vx + x}{2vx - x} = \frac{v + 1}{2v - 1}$$

$\therefore x\frac{dv}{dx} = \frac{v+1}{2v-1} - v = \frac{v + 1 - 2v^2 + v}{2v - 1}$

or $\frac{2v - 1}{-2v^2 + 2v + 1}dv = \frac{dx}{x}$ on solving, we get

$$\sin^2 y = x\sin y + \frac{x^2}{2} + c.$$

**Ans. (A)**



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***Illustration 18 :*** Solve the differential equation $(1+2e^{x/y})\,dx + 2e^{x/y}(1-x/y)\,dy = 0$.

***Solution :*** The equation is homogeneous of degree 0.

Put $x = vy$, $dx = v\,dy + y\,dv$,

Then, differential equation becomes

$(1+2e^{v})(v\,dy + y\,dv) + 2e^{v}(1-v)\,dy = 0 \Rightarrow (v+2e^{v})\,dy + y(1+2e^{v})\,dv = 0$

$$\frac{dy}{y} + \frac{1+2e^{v}}{v+2e^{v}}\,dv = 0$$

Integrating and replacing v by x/y, we get $\ell n\, y + \ell n(v+2e^{v}) = \ell nc$ and $x + 2ye^{x/y} = c$

**Ans.**

**Do yourself - 9 :**

Solve the following differential equations :

**1.** $y' = \dfrac{3x-y}{x+y}$

**2.** $(x - y\ell ny + y\ell nx)\,dx + x(\ell ny - \ell nx)\,dy = 0$

**3.** $(3xy + y^2)dx + (x^2 + xy)dy = 0,\ y(1) = 1$

**4.** $\dfrac{dy}{dx} = \dfrac{x^2+xy}{x^2+y^2}$

**(ii) Equations reducible to homgeneous form**

The equation of the form $\dfrac{dy}{dx} = \dfrac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ where $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$

can be reduced to homgeneous form by changing the variable x, y to u,v as

$x = u + h$ , $y = v + k$

where h,k are the constants to be chosen so as to make the given equation homgeneous.

We have

$$\frac{dy}{dx} = \frac{dv}{du}$$

$\therefore$ The equation becomes, $\dfrac{dv}{du} = \dfrac{a_1u + b_1v + (a_1h + b_1k + c_1)}{a_2u + b_2v + (a_2h + b_2k + c_2)}$

Let h and k be chosen so as to satisfy the equation

$a_1h + b_1k + c_1 = 0$ ...(i)

$a_2h + b_2k + c_2 = 0$ ...(ii)

Solve for h and k from (i) and (ii)

Now $\dfrac{du}{dv} = \dfrac{a_1u + b_1v}{a_2u + b_2v}$

is a homgeneous equation and can be solved by substituting $v = ut$.

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***Illustration 19 :*** Solve $\frac{dy}{dx} = \frac{x+2y+3}{2x+3y+4}$

***Solution :*** Put $x = X + h$, $y = Y + k$

We have $\frac{dY}{dX} = \frac{X+2Y+(h+2k+3)}{2X+3Y+(2h+3k+4)}$

To determine h and k, we write

$h + 2k + 3 = 0$, $2h + 3k + 4 = 0 \Rightarrow h = 1, k = -2$

So that $\frac{dY}{dX} = \frac{X+2Y}{2X+3Y}$

Putting $Y = VX$, we get

$$V + X\frac{dV}{dX} = \frac{1+2V}{2+3V} \Rightarrow \frac{2+3V}{3V^2-1}dV = -\frac{dX}{X}$$

$$\Rightarrow \left[\frac{2+\sqrt{3}}{2(\sqrt{3}V-1)} - \frac{2-\sqrt{3}}{2(\sqrt{3}V+1)}\right]dV = -\frac{dX}{X}$$

$$\Rightarrow \frac{2+\sqrt{3}}{2\sqrt{3}}\log(\sqrt{3}V - 1) - \frac{2-\sqrt{3}}{2\sqrt{3}}\log(\sqrt{3}V + 1) = (-\log X + c)$$

$$\frac{2+\sqrt{3}}{2\sqrt{3}}\log(\sqrt{3}Y - X) - \frac{2-\sqrt{3}}{2\sqrt{3}}\log(\sqrt{3}Y + X) = A$$ where A is another constant

and $X = x - 1$, $Y = y + 2$. **Ans.**

**Do yourself - 10 :**

**1.** Solve the differential equation : $\frac{dy}{dx} = \frac{x+2y-5}{2x+y-4}$

**2.** $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

**3.** $\frac{dy}{dx} = \frac{y-x+1}{y+x+5}$

**4.** $\frac{dy}{dx} = \frac{x+y+1}{2x+2y+3}$

**5.** $(x - y)dy = (x + y + 1)dx$

**(c) Linear differential equations :**

A differential equation is said to be linear if the dependent variable & its differential coefficients occur in the first degree only and are not multiplied together.

The $n^{th}$ order linear differential equation is of the form ;

$a_0(x)\frac{d^ny}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \ldots\ldots + a_n(x)\,y = \phi(x)$, where $a_0(x)$, $a_1(x)$ .... $a_n(x)$ are called the coefficients of the differential equation.

Note that a linear differential equation is always of the first degree but every differential equation of the first degree need not be linear . e.g. the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y^2 = 0$ is not linear, though its degree is 1.

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***Illustration 20 :*** Which of the following equation is linear ?

(A) $\frac{dy}{dx} + xy^2 = 1$ (B) $x^2\frac{dy}{dx} + y = e^x$ (C) $\frac{dy}{dx} + 3y = xy^2$ (D) $x\frac{dy}{dx} + y^2 = \sin x$

***Solution :*** Clearly answer is (B)

***Illustration 21 :*** Which of the following equation is non-linear ?

(A) $\frac{dy}{dx} = \cos x$ (B) $\frac{d^2y}{dx^2} + y = 0$ (C) $dx + dy = 0$ (D) $x\left(\frac{dy}{dx}\right) + \frac{3}{\left(\frac{dy}{dx}\right)} = y^2$

***Solution :*** Clearly answer is (D)

**(i) Linear differential equations of first order :**

The most general form of a linear differential equation of first order is $\frac{dy}{dx} + Py = Q$, where P & Q are functions of x.

To solve such an equation multiply both sides by $e^{\int Pdx}$.

So that we get $e^{\int Pdx}\left[\frac{dy}{dx} + Py\right] = Qe^{\int Pdx}$ ....(i)

$$\frac{d}{dx}\left(e^{\int Pdx}.y\right) = Qe^{\int Pdx} \quad ...(ii)$$

On integrating equation (ii), we get

$$ye^{\int Pdx} = \int Qe^{\int Pdx}dx + c$$

This is the required general solution.

**Note :**

(i) The factor $e^{\int Pdx}$ on multiplying by which the left hand side of the differential equation becomes the differential coefficient of some function of x & y, is called integrating factor of the differential equation popularly abbreviated as I.F.

(ii) Sometimes a given differential equation becomes linear if we take y as the independent variable and x as the dependent variable. e.g. the equation ; $(x + y + 1)\frac{dy}{dx} = y^2 + 3$ can be written as $(y^2 + 3)\frac{dx}{dy} = x + y + 1$ which is a linear differential equation.

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***Illustration 22 :*** Solve $(1+y^2)+(x-e^{\tan^{-1}y})\dfrac{dy}{dx}=0$.

***Solution :*** Differential equation can be rewritten as $(1+y^2)\dfrac{dx}{dy}+x=e^{\tan^{-1}y}$

or $\dfrac{dx}{dy}+\dfrac{1}{1+y^2}.x=\dfrac{e^{\tan^{-1}y}}{1+y^2}$ ...(i)

I. F $=e^{\int\frac{1}{1+y^2}dy}=e^{\tan^{-1}y}$

so solution is $xe^{\tan^{-1}y}=\int\dfrac{e^{\tan^{-1}y}e^{\tan^{-1}y}}{1+y^2}dy$

Let $e^{\tan^{-1}y}=t\Rightarrow\dfrac{e^{\tan^{-1}y}}{1+y^2}dy=dt$

$xe^{\tan^{-1}y}=\int t\,dt$ [Putting $e^{\tan^{-1}y}=t$]

or $xe^{\tan^{-1}y}=\dfrac{t^2}{2}+\dfrac{c}{2}\Rightarrow 2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+c.$ **Ans.**

***Illustration 23 :*** The solution of differential equation $(x^2-1)\dfrac{dy}{dx}+2xy=\dfrac{1}{x^2-1}$ is -

(A) $y(x^2-1)=\dfrac{1}{2}\log\left|\dfrac{x-1}{x+1}\right|+C$ (B) $y(x^2+1)=\dfrac{1}{2}\log\left|\dfrac{x-1}{x+1}\right|-C$

(C) $y(x^2-1)=\dfrac{5}{2}\log\left|\dfrac{x-1}{x+1}\right|+C$ (D) none of these

***Solution :*** The given differential equation is

$(x^2-1)\dfrac{dy}{dx}+2\ xy=\dfrac{1}{x^2-1}\Rightarrow\dfrac{dy}{dx}+\dfrac{2x}{x^2-1}y=\dfrac{1}{(x^2-1)^2}$ ...(i)

This is linear differential equation of the form

$\dfrac{dy}{dx}+Py=Q,$ where $P=\dfrac{2x}{x^2-1}$ and $Q=\dfrac{1}{(x^2-1)^2}$

$\therefore$ I.F. $=e^{\int P\,dx}=e^{\int 2x/(x^2-1)dx}=e^{\log(x^2-1)}=(x^2-1)$

multiplying both sides of (i) by I.F. $=(x^2-1)$, we get

$(x^2-1)\dfrac{dy}{dx}+2\ xy=\dfrac{1}{x^2-1}$

integrating both sides we get

$y(x^2-1)=\int\dfrac{1}{x^2-1}dx+C$ [Using : $y(\text{I.F.})=\int Q.(\text{I.F.})\ dx+C$]

$\Rightarrow\ y(x^2-1)=\dfrac{1}{2}\log\left|\dfrac{x-1}{x+1}\right|+C.$

This is the required solution. **Ans.(A)**

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**Do yourself - 11 :**

Solve the following differential equations :

**1.** $\frac{xdy}{dx} = 2y + x^4 + 6x^2 + 2x,\ x \neq 0$

**2.** $(x-a)\frac{dy}{dx} + 3y = 12(x-a)^3,\ x > a > 0$

**3.** $y\,\ell n\,y\,dx + (x - \ell n\,y)\,dy = 0$

**4.** $\frac{dy}{dx} + \frac{x}{1+x^2}y = \frac{1}{2x(1+x^2)}$

**5.** $(1-x^2)\frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$

**6.** $x(x-1)\frac{dy}{dx} - (x-2)y = x^3(2x-1)$

**(ii) Equation reducible to linear form :**

The equation of the form $\frac{dy}{dx} + Py = Qy^n$, where P and Q are functions of x, is called **Bernoulli's equation.**

On dividing by $y^n$, we get $y^{-n}\frac{dy}{dx} + Py^{-n+1} = Q$

Let $y^{-n+1} = t$, so that $(-n+1)y^{-n}\frac{dy}{dx} = \frac{dt}{dx}$

then equation becomes $\frac{dt}{dx} + P(1-n)t = Q(1-n)$

which is linear with t as a dependent variable.

---

***Illustration 24 :*** Solve the differential equation $x\frac{dy}{dx} + y = x^3y^6$.

***Solution :*** The given differential equation can be written as

$$\frac{1}{y^6}\frac{dy}{dx} + \frac{1}{xy^5} = x^2$$

Putting $y^{-5} = v$ so that

$$-5y^{-6}\frac{dy}{dx} = \frac{dv}{dx} \text{ or } y^{-6}\frac{dy}{dx} = -\frac{1}{5}\frac{dv}{dx} \text{ we get}$$

$$-\frac{1}{5}\frac{dv}{dx} + \frac{1}{x}v = x^2 \Rightarrow \frac{dv}{dx} - \frac{5}{x}v = -5x^2 \quad \text{.....(i)}$$

This is the standard form of the linear deferential equation having integrating factor

$$\text{I.F} = e^{\int -\frac{5}{x}dx} = e^{-5\log x} = \frac{1}{x^5}$$

Multiplying both sides of (i) by I.F. and integrating w.r.t. x

We get $v\cdot\frac{1}{x^5} = \int -5x^2\cdot\frac{1}{x^5}dx$

$$\Rightarrow \frac{v}{x^5} = \frac{5}{2}x^{-2} + c$$

$\Rightarrow y^{-5}x^{-5} = \frac{5}{2}x^{-2} + c$ which is the required solution. **Ans.**



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***Illustration 25 :*** Find the solution of differential equation $\frac{dy}{dx} - y \tan x = -y^2 \sec x$.

***Solution :*** $\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y}\tan x = -\sec x$

$\frac{1}{y} = v$ ; $\frac{-1}{y^2}\frac{dy}{dx} = \frac{dv}{dx}$ $\quad\therefore\ \frac{-dv}{dx} - v\tan x = -\sec x$

$\frac{dv}{dx} + v\tan x = \sec x$, $\quad$ Here $P = \tan x$, $Q = \sec x$

I.F. $= e^{\int \tan x dx} = |\sec x|$ $\quad v|\sec x| = \int \sec^2 x\, dx + c$

Hence the solution is $y^{-1}|\sec x| = \tan x + c$ **Ans.**

**Do yourself - 12 :**

Solve the following differential equations :

**1.** $y' + 3y = e^{3x} y^2$

**2.** $xdy - \{y + xy^3(1 + \ell nx)\}dx = 0$

**3.** $\frac{dy}{dx} + y = y^2(\cos x - \sin x)$

**4.** $2\frac{dy}{dx} - y\sec x = y^3 \tan x$

**5.** $x^2 y - x^3\frac{dy}{dx} = y^4 \cos x$

**6.** $y(2xy + e^x)\,dx - e^x\,dy = 0$

**7.** $\sin x\frac{dy}{dx} + 3y = \cos x$

**8.** $x(x^2 + 1)\frac{dy}{dx} = y(1 - x^2) + x^3 . \ell nx$

**9.** $x\frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

**10.** $(1 + y^2)\,dx = (\tan^{-1} y - x)dy$

**11.** $(x - y^2)\,dx + 2xy\,dy = 0$

**12.** $(x^3 + y^2 + 2)\,dx + 2y\,dy = 0$

**13.** $x\frac{dy}{dx} + y\ln y = xye^x$

**14.** $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1 + x)\,e^x \sec y$

**15.** $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$

**16.** $\left(\frac{dy}{dx}\right)^2 - (x+y)\frac{dy}{dx} + xy = 0$

**17.** $\frac{dy}{dx} = \frac{y^2 - x}{2y(x + 1)}$

**18.** $(1 - xy + x^2y^2)\,dx = x^2\,dy$

**19.** $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

**20.** $(x^2 + y)\frac{dy}{dx} = 6x$

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## 7. TRAJECTORIES :

A curve which cuts every member of a given family of curves according to a given law is called a Trajectory of the given family.

The trajectory will be called **Orthogonal** if each trajectory cuts every member of given family at right angle.

**Working rule for finding orthogonal trajectory**

**1.** Form the differential equation of family of curves

**2.** Write $-\frac{1}{dy/dx}$ for $\frac{dy}{dx}$ or $-\frac{r^2 d\theta}{dr}$ for $\frac{dr}{d\theta}$ if differential equation is in the polar form.

**3.** Solve the new differential equation to get the equation of orthogonal trajectories.

**Note :** A family of curves is self-orthogonal if it is its own orthogonal family.

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***Illustration 26 :*** Find the value of k such that the family of parabolas $y = cx^2 + k$ is the orthogonal trajectory of the family of ellipses $x^2 + 2y^2 - y = c$.

***Solution :*** Differentiate both sides of $x^2 + 2y^2 - y = c$ w.r.t. x, We get

$$2x + 4y\frac{dy}{dx} - \frac{dy}{dx} = 0$$

or $2x + (4y - 1)\frac{dy}{dx} = 0$, is the differential equation of the given family of curves.

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ to obtain the differential equation of the orthogonal trajectories, we get

$$2x + \frac{(1-4y)}{\frac{dy}{dx}} = 0 \Rightarrow \frac{dy}{dx} = \frac{4y-1}{2x}$$

$$\Rightarrow \int \frac{dy}{4y-1} = \int \frac{dx}{2x} \Rightarrow \frac{1}{4}\ell n\,(4y-1) = \frac{1}{2}\ell n\, x + \frac{1}{2}\ell n\, a, \text{ where a is any constant.}$$

$$\Rightarrow \ell n(4y-1) = 2\,\ell n\, x + 2\,\ell n\, a \text{ or }, 4y - 1 = a^2x^2$$

or, $y = \frac{1}{4}a^2x^2 + \frac{1}{4}$, is the required orthogonal trajectory, which is of the form

$y = cx^2 + k$ where $c = \frac{a^2}{4}$, $k = \frac{1}{4}$. **Ans.**

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***Illustration 27 :*** Prove that $\frac{x^2}{1+\lambda}+\frac{y^2}{4+\lambda}=1$ are self orthogonal family of curves.

***Solution :*** $\frac{x^2}{1+\lambda}+\frac{y^2}{4+\lambda}=1$ ...(i)

Differentiating (i) with respect to x, we have

$\frac{x}{1+\lambda}+\frac{y}{4+\lambda}\frac{dy}{dx}=0$ ...(ii)

From (i) and (ii), we have to eliminate $\lambda$.

Now, (ii) gives

$$\lambda=\frac{-\left[4x+y\frac{dy}{dx}\right]}{x+y\frac{dy}{dx}}$$

$$\Rightarrow 1+\lambda=\frac{(1-4)x}{x+y(dy/dx)},\quad 4+\lambda=\frac{-(1-4)y(dy/dx)}{x+y(dy/dx)}$$

Substituting these values in (i), we get

$\left(x+y\frac{dy}{dx}\right)\left(x-y\frac{dx}{dy}\right)=-3$ ....(iii)

as the differential equation of the given family.

Changing dy/dx to –dx/dy in (iii), we obtain

$\left(x-y\frac{dx}{dy}\right)\left(x+y\frac{dy}{dx}\right)=-3$ ...(iv)

which is the same as (iii). Thus we see that the family (i) as self-orthogonal, i.e., every member of the family (i) cuts every other member of the same family orthogonally.

**Do yourself - 13 :**

**1.** Find the orthogonal trajectories of the following families of curves :

**(a)** $x+2y=C$ **(b)** $y=Ce^{-2x}$

**2.** Find the orthogonal trajectories for the given family of curves when 'a' is the parameter.

**(a)** $y=ax^2$ **(b)** $\cos y=a\,e^{-x}$

**Note :**

Following exact differentials must be remembered :

(i) $xdy+y\,dx=d(xy)$ (ii) $\frac{xdy-ydx}{x^2}=d\left(\frac{y}{x}\right)$ (iii) $\frac{ydx-xdy}{y^2}=d\left(\frac{x}{y}\right)$

(iv) $\frac{xdy+ydx}{xy}=d\left(\ell nxy\right)$ (v) $\frac{dx+dy}{x+y}=d\left(\ell n(x+y)\right)$ (vi) $\frac{xdy-ydx}{xy}=d\left(\ell n\frac{y}{x}\right)$

(vii) $\frac{ydx-xdy}{xy}=d\left(\ell n\frac{x}{y}\right)$ (viii) $\frac{xdy-ydx}{x^2+y^2}=d\left(\tan^{-1}\frac{y}{x}\right)$ (ix) $\frac{ydx-xdy}{x^2+y^2}=d\left(\tan^{-1}\frac{x}{y}\right)$

(x) $\frac{xdx+ydy}{x^2+y^2}=d\left[\ell n\sqrt{x^2+y^2}\right]$ (xi) $d\left(-\frac{1}{xy}\right)=\frac{xdy+ydx}{x^2y^2}$ (xii) $d\left(\frac{e^x}{y}\right)=\frac{ye^xdx-e^xdy}{y^2}$

(xiii) $d\left(\frac{e^y}{x}\right)=\frac{xe^ydy-e^ydx}{x^2}$

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***Illustration 28 :*** Solve $\dfrac{y+\sin x\cos^2(xy)}{\cos^2(xy)}dx+\left(\dfrac{x}{\cos^2(xy)}+\sin y\right)dy=0.$

***Solution :*** The given differential equation can be written as;

$$\frac{y\,dx+x\,dy}{\cos^2(xy)}+\sin x\,dx+\sin y\,dy=0.$$

$\Rightarrow \sec^2(xy)\,d(xy)+\sin x\,dx+\sin y\,dy=0$

$\Rightarrow d(\tan(xy))+d(-\cos x)+d(-\cos y)=0$

$\Rightarrow \tan(xy)-\cos x-\cos y=c.$ **Ans.**

**Do yourself - 14 :**

Solve the following differential equations :

**1.** $x\,dx+y\,dy+4y^3(x^2+y^2)dy=0.$ **2.** $x\,dy-ydx-(1-x^2)dx=0.$

## 8. APPLICATION OF DIFFERENTIAL EQUATIONS :

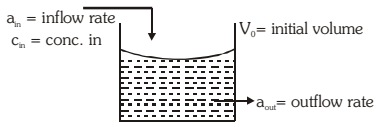
### (a) Mixing Problems

A chemical in a liquid solution with given concentration $c_{in}$ gm/lit. (or dispersed in a gas) runs into a container with a rate of $a_{in}$ lit/min. holding the liquid (or the gas) with, possibly, a specified amount of the chemical dissolved as well. The mixture is kept uniform by stirring and flows out of the container at a known rate ($a_{out}$ litre/min.). In this process it is often important to know the concentration of the chemical in the container at any given time. The differential equation describing the process is based on the formula.

$$\begin{pmatrix}\text{Rate of change}\\ \text{of amount}\\ \text{in container}\end{pmatrix}=\begin{pmatrix}\text{rate at which}\\ \text{chemical}\\ \text{arrives}\end{pmatrix}-\begin{pmatrix}\text{rate at which}\\ \text{chemical}\\ \text{departs}\end{pmatrix} \qquad \text{.....(i)}$$

Arrival rate = (conc. in) × (inflow rate) = $c_{in}\times a_{in}$

If y(t) denotes the amount of substance in the tank at time t & V(t) denotes the amount of mixture in tank at that time




$$\text{Departure rate}=\begin{pmatrix}\text{concentration in}\\ \text{container at time t}\end{pmatrix}.(\text{outflow rate})$$

$$=\frac{y(t)}{V(t)}.(a_{out})$$

where volume of mixture at time t, V(t) = initial volume + (inflow rate – outflow rate) × t

$$= V_0+(a_{in}-a_{out})t$$

Accordingly, Equation (i) becomes

$$\frac{dy(t)}{dt}=(\text{chemical's given arrival rate})-\frac{y(t)}{V(t)}\cdot(\text{out flow rate}) \qquad \text{.....(ii)}$$

$$\frac{d(y(t))}{dt}=c_{in}a_{in}-\frac{y(t)}{V_0+(a_{in}-a_{out})t}.a_{out}$$

This leads to a first order linear D.E. which can be solved to obtain y(t) i.e. amount of chemical at time 't'.

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***Illustration 29 :*** A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains 0.03 kg of salt per liter of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt remains in the tank after half an hour ?

***Solution :*** Let y(t) be the amount of salt after t min.

Given $y(0) = 20$ kg

$$\text{rate in} = \left(\frac{0.03\text{kg}}{\text{L}}\right)\left(\frac{25\text{L}}{\text{min.}}\right) = \frac{0.75\text{kg}}{\text{min.}}$$

As $a_{in} = a_{out}$, so the tank always contains 5000 L of liquid so the conc. at time 't' is $\left(\frac{y(t)}{5000}\right)\frac{\text{kg}}{\text{L}}$

so $$\text{rate out} = \left(\frac{y(t)}{5000}\frac{\text{kg}}{\text{L}}\right)\left(\frac{25\text{L}}{\text{min}}\right) = \frac{y(t)}{200}\frac{\text{kg}}{\text{min}}$$

$$\frac{dy(t)}{dt} = 0.75 - \frac{y(t)}{200}$$

by solving as linear D.E. or variable separable and using initial condition, we get

$$y(t) = 150 - 130\, e^{-t/200}$$

The amount of salt after 30 min is

$$y(30) = 150 - 130\, e^{-30/100} = 38.1 \text{ kg}$$

**Do yourself - 15 :**

**1.** A tank initially holds 10 lit. of fresh water. At t = 0, a brine solution containing $\frac{1}{2}$ kg of salt per lit. is poured into the tank at a rate of 2 lit/min. while the well-stirred mixture leaves the tank at the same rate. Find

**(a)** the amount and **(b)** the concentration of salt in the tank at any time t.

**2.** A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min and the mixture in pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

**3.** A tank consists of 50 litres of fresh water. Two litres of brine each litre containing 5 gms of dissolved salt are run into tank per minute; the mixture is kept uniform by stirring, and runs out at the rate of one litre per minute. If 'm' grams of salt are present in the tank after t minute, express 'm' in terms of t and find the amount of salt present after 10 minutes.

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**(b) Exponential Growth and Decay :**

In general, if y(t) is the value of quantity y at time t and if the rate of change of y with respect to t is proportional to its value y(t) at that time, then

$$\frac{dy(t)}{dt} = ky(t), \text{ where k is a constant} \qquad ....(i)$$

$$\int \frac{dy(t)}{y(t)} = \int k\,dt$$

Solving, we get $y(t) = Ae^{kt}$

equation (i) is sometimes called the law of natural growth (if $k > 0$) or law of natural decay (if $k < 0$).

In the context of population growth, we can write

$$\frac{dP}{dt} = kP \quad \text{or} \quad \frac{1}{P}\frac{dP}{dt} = k$$

where k is growth rate divided by the population size; it is called **the relative growth rate.**

---

***Illustration 30 :*** A certain radioactive material is known to decay at a rate proportional to the amount present. If initially there is 50 kg of the material present and after two hours it is observed that the material has lost 10 percent of its original mass, find **(a)** an expression for the mass of the material remaining at any time t, **(b)** the mass of the material after four hours, and **(c)** the time at which the material has decayed to one half of its initial mass.

***Solution :*** **(a)** Let N denote the amount of material present at time t.

$$\text{So, } \frac{dN}{dt} - kN = 0$$

This differential equation is separable and linear, its solution is

$$N = ce^{kt} \qquad ....(i)$$

At $t = 0$, we are given that $N = 50$. Therefore, from (i), $50 = ce^{k(0)}$ or $c = 50$. Thus,

$$N = 50e^{kt} \qquad ....(ii)$$

At $t = 2$, 10 percent of the original mass of 50kg or 5kg has decayed. Hence, at $t = 2$, $N = 50 - 5 = 45$. Substituting these values into (ii) and solving for k, we have

$$45 = 50e^{2k} \text{ or } k = \frac{1}{2}\ell n\,\frac{45}{50}$$

Substituting this value into (ii), we obtain the amount of mass present at any time t as

$$N = 50e^{\frac{1}{2}(\ell n\,0.9)t} \qquad .....(iii)$$

where t is measured in hours.

**(b)** We require N at $t = 4$. Substituting $t = 4$ into (iii) and then solving for N, we find

$N = 50e^{-2\,\ell n\,(0.9)}$ kg

**(c)** We require when $N = 50/2 = 25$. Substituting $N = 25$ into (iii) and solving for t, we find

$$25 = 50e^{\frac{1}{2}(\ell n\,0.9)t} \Rightarrow t = \ell n\left(\frac{1}{2}\right)\Big/\left[\frac{1}{2}\ell n\,(0.9)\right] \text{ hours}$$

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**(c) Geometrical applications :**

Let $P(x_1, y_1)$ be any point on the curve $y = f(x)$, then slope of the tangent at point P is $\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$

(i) The equation of the tangent at P is $y - y_1 = \frac{dy}{dx}(x - x_1)$

x-intercept of the tangent $= x_1 - y_1\left(\frac{dx}{dy}\right)$

y-intercept of the tangent $= y_1 - x_1\left(\frac{dy}{dx}\right)$

(ii) The equation of normal at P is $y - y_1 = -\frac{1}{(dy/dx)}(x - x_1)$

x and y-intercepts of normal are ; $x_1 + y_1\frac{dy}{dx}$ and $y_1 + x_1\frac{dx}{dy}$

(iii) Length of tangent $= PT = |y_1|\sqrt{1 + (dx/dy)^2_{(x_1, y_1)}}$

(iv) Length of normal $= PN = |y_1|\sqrt{1 + (dy/dx)^2_{(x_1, y_1)}}$

(v) Length of sub-tangent $= ST = \left|y_1\left(\frac{dx}{dy}\right)_{(x_1, y_1)}\right|$

(vi) Length of sub-normal $= SN = \left|y_1\left(\frac{dy}{dx}\right)_{(x_1, y_1)}\right|$

(vii) Length of radius vector $= \sqrt{x_1^2 + y_1^2}$

**Do yourself - 16 :**

**1.** At each point (x,y) of a curve the intercept of the tangent on the y-axis is equal to $2xy^2$. Find the curve.

**2.** Find the equation of the curve for which the normal at any point (x,y) passes through the origin.

**3.** It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was $m_0$ and during time $t_0$ $\alpha$ % of the original mass of radium decay.

**4.** A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

**5.** The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

**(a)** Population at any time t, given initial population of the town being 2500.

**(b)** If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

**(c)** Predict about the population of the town in the long run.

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***Miscellaneous Illustrations :***

***Illustration 31 :*** Solve $(y \log x - 1)\, ydx = xdy$.

***Solution :*** The given differential equation can be written as

$$x\frac{dy}{dx} + y = y^2 \log x \qquad \text{.....(i)}$$

Divide by $xy^2$. Hence $\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x}\log x$

Let $\frac{1}{y} = v \Rightarrow -\frac{1}{y^2}\frac{dy}{dx} = \frac{dv}{dx}$ so that $\frac{dv}{dx} - \frac{1}{x}v = -\frac{1}{x}\log x$ .....(ii)

(ii) is the standard linear differential equation with $P = -\frac{1}{x}$, $Q = -\frac{1}{x}\log x$

I.F. $= e^{\int pdx} = e^{\int -1/x\, dx} = 1/x$

The solution is given by

$$v \cdot \frac{1}{x} = \int \frac{1}{x}\left(-\frac{1}{x}\log x\right)dx = -\int \frac{\log x}{x^2}dx = \frac{\log x}{x} - \int \frac{1}{x}\cdot\frac{1}{x}dx = \frac{\log x}{x} + \frac{1}{x} + C$$

$\Rightarrow \quad v = 1 + \log x + cx = \log ex + cx$

or $\quad \frac{1}{y} = \log ex + cx$ or $y(\log ex + cx) = 1$. **Ans.**

***Illustration 32:*** For a certain curve $y = f(x)$ satisfying $\frac{d^2y}{dx^2} = 6x - 4$, $f(x)$ has a local minimum value 5 when $x = 1$.

Find the equation of the curve and also the global maximum and global minimum values of $f(x)$ given that $0 \le x \le 2$.

***Solution :*** Integrating $\frac{d^2y}{dx^2} = 6x - 4$, we get $\frac{dy}{dx} = 3x^2 - 4x + A$

When $x = 1$, $\frac{dy}{dx} = 0$, so that $A = 1$. Hence

$$\frac{dy}{dx} = 3x^2 - 4x + 1 \qquad \text{...(i)}$$

Integrating, we get $y = x^3 - 2x^2 + x + B$

When $x = 1$, $y = 5$, so that $B = 5$.

Thus we have $y = x^3 - 2x^2 + x + 5$.

From (i), we get the critical points $x = 1/3$, $x = 1$

At the critical point $x = \frac{1}{3}$, $\frac{d^2y}{dx^2}$ is negative.

Therefore at $x = 1/3$, $y$ has a local maximum.

At $x = 1$, $\frac{d^2y}{dx^2}$ is positive.

Therefore at $x = 1$, $y$ has a local minimum.

Also $f(1) = 5$, $f\left(\frac{1}{3}\right) = \frac{139}{27}$. $f(0) = 5$, $f(2) = 7$

Hence the global maximum value = 7, and the global minimum value = 5. **Ans.**

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***Illustration 33 :*** Solve $\frac{dy}{dx} = \tan y \cot x - \sec y \cos x.$

***Solution :*** $\frac{dy}{dx} = \tan y \cot x - \sec y \cos x.$

Rearrange it :

$(\sin x - \sin y)\cos x\, dx + \sin x \cos y\, dy = 0.$

Put $u = \sin y$, So, $du = \cos y\, dy$ :

Substituting, we get

$$(\sin x - u)\cos x\, dx + \sin x\, du = 0,\ \frac{du}{dx} - u\frac{\cos x}{\sin x} = -\cos x$$

The equation is first-order linear in u.

The integrating factor is

$$I = \exp\int -\frac{\cos x}{\sin x}dx = \exp\{-\ln(\sin x)\} = \frac{1}{\sin x}.$$

Hence, $$u\frac{1}{\sin x} = -\int \frac{\cos x}{\sin x}dx = -\ln|\sin x| + C.$$

Solve for u : $u = -\sin x \ln|\sin x| + C \sin x.$

Put y back : $\sin y = -\sin x \ln|\sin x| + C \sin x.$ **Ans.**

***Illustration 34 :*** Solve the equation $x\int_0^x y(t)dt = (x+1)\int_0^x t\,y(t)dt, x > 0$

***Solution :*** Differentiating the equation with respect to x, we get

$$xy(x) + 1.\int_0^x y(t)dt = (x+1)xy(x) + 1.\int_0^x ty(t)dt$$

i.e., $$\int_0^x y(t)dt = x^2y(x) + \int_0^x ty(t)dt$$

Differentiating again with respect to x, we get $y(x) = x^2 y'(x) + 2xy(x) + xy(x)$

i.e., $$(1-3x)y(x) = \frac{x^2dy(x)}{dx}$$

i.e., $$\frac{(1-3x)dx}{x^2} = \frac{dy(x)}{y(x)},$$ integrating we get

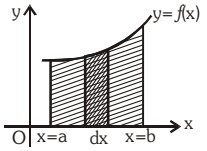
i.e., $$y = \frac{c}{x^3}e^{-1/x}$$ **Ans.**



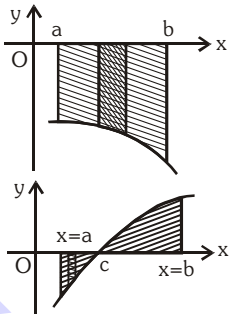
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# AREA UNDER THE CURVE

## 1. AREA UNDER THE CURVES :

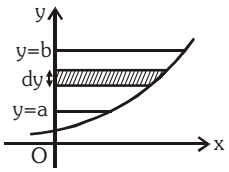



**(a)** Area bounded by the curve $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$ is given by $A = \int_a^b y\,dx$ , where $y = f(x)$ lies above the x-axis and $b > a$. Here vertical strip of thickness dx is considered at distance x.




**(b)** If $y = f(x)$ lies completely below the x-axis then A is negative and we consider the magnitude only, i.e. $A = \left|\int_a^b y\,dx\right|$

**(c)** If curve crosses the x-axis at $x = c$, then $A = \left|\int_a^c y\,dx\right| + \int_c^b y\,dx$




**(d)** Sometimes integration w.r.t. y is very useful (horizontal strip) :

Area bounded by the curve, y-axis and the two abscissae at $y = a$ & $y = b$ is written as $A = \int_a^b x\,dy$ .

**Note :** If the curve is symmetric and suppose it has 'n' symmetric portions, then total area = n (Area of one symmetric portion).

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***Illustration 1 :*** Find the area bounded by $y = \sec^2 x$, $x = \frac{\pi}{6}$, $x = \frac{\pi}{3}$ & x-axis

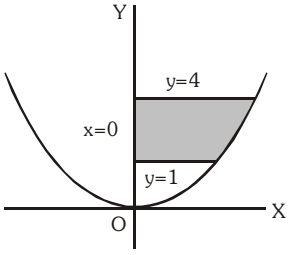
***Solution :*** $\text{Area bounded} = \int_{\pi/6}^{\pi/3} y\,dx = \int_{\pi/6}^{\pi/3} \sec^2 x\,dx = [\tan x]_{\pi/6}^{\pi/3} = \tan\frac{\pi}{3} - \tan\frac{\pi}{6} = \sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$ sq.units.

***Illustration 2 :*** Find the area in the first quadrant bounded by $y = 4x^2$, $x = 0$, $y = 1$ and $y = 4$.

***Solution :*** Required area $= \int_1^4 x\,dy = \int_1^4 \frac{\sqrt{y}}{2}\,dy = \frac{1}{2}\left[\frac{2}{3}y^{3/2}\right]_1^4$

$= \frac{1}{3}[4^{3/2} - 1] = \frac{1}{3}[8 - 1]$

$= \frac{7}{3} = 2\frac{1}{3}$ sq.units.

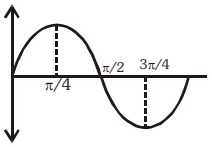



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***Illustration 3 :*** Find the area bounded by the curve $y = \sin 2x$, x-axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

**Solution :** Required area $= \int_{\pi/4}^{\pi/2} \sin 2x\,dx + \left|\int_{\pi/2}^{3\pi/4} \sin 2x\,dx\right| = \left(-\frac{\cos 2x}{2}\right)\Big|_{\pi/4}^{\pi/2} + \left|\left(-\frac{\cos 2x}{2}\right)\Big|_{\pi/2}^{3\pi/4}\right|$

$= -\frac{1}{2}[-1-0] + \left|\frac{1}{2}(0+(-1))\right| = 1$ sq. unit




**Do yourself - 1 :**

**1.** Find the area bounded by $y = x^2 + 2$ above x-axis between $x = 2$ & $x = 3$.

**2.** Using integration, find the area of the curve $y = \sqrt{1-x^2}$ with co-ordinate axes bounded in first quadrant.

**3.** Find the area bounded by the curve $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$.

**4.** Find the area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = -\frac{1}{2}$ and $x = 1$.

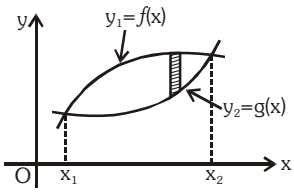
**5.** Find the area bounded by $x = 2$, $x = 5$, $y = x^2$, $y = 0$.

**6.** Find the area enclosed between $y = \cos x$ & x-axis as x varies from $\frac{-\pi}{2}$ to $\frac{3\pi}{2}$

**7.** Find area enclosed between $y = \ell nx$, x-axis and the two ordinates $x = \frac{1}{e^2}$ and $x = e^2$.

**8.** Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes

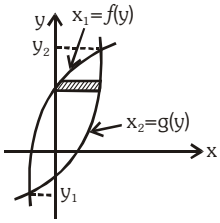
## 2. AREA ENCLOSED BETWEEN TWO CURVES :




**(a)** Area bounded by two curves $y = f(x)$ & $y = g(x)$ such that $f(x) > g(x)$ is

$$A = \int_{x_1}^{x_2} (y_1 - y_2)\,dy$$

$$A = \int_{x_1}^{x_2} [f(x) - g(x)]\,dx$$




**(b)** In case horizontal strip is taken we have

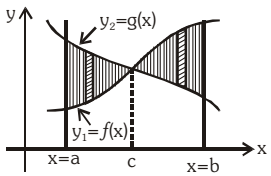
$$A = \int_{y_1}^{y_2} (x_1 - x_2)\,dy$$

$$A = \int_{y_1}^{y_2} [f(y) - g(y)]\,dy$$

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**(c)** If the curves $y_1 = f(x)$ and $y_2 = g(x)$ intersect at $x = c$, then required area

$$A = \int_a^c (g(x) - f(x))dx + \int_c^b (f(x) - g(x))dx = \int_a^b |f(x) - g(x)|dx$$




**Note :** Required area must have all the boundaries indicated in the problem.

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***Illustration 4 :*** Find the area bounded by the curve $y = (x-1)(x-2)(x-3)$ lying between the ordinates $x = 0$ and $x = 3$ and x-axis

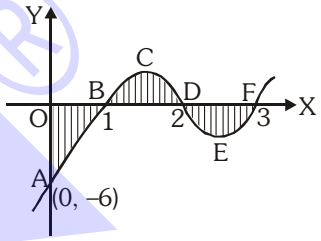
***Solution :*** To determine the sign, we follow the usual rule of change of sign.

$y = +ve$ for $x > 3$

$y = -ve$ for $2 < x < 3$

$y = +ve$ for $1 < x < 2$

$y = -ve$ for $x < 1$.




$$\int_0^3 |y|\,dx = \int_0^1 |y|\,dx + \int_1^2 |y|\,dx + \int_2^3 |y|\,dx$$

$$= \int_0^1 -y\,dx + \int_1^2 y\,dx + \int_2^3 -y\,dx$$

Now let $F(x) = \int (x-1)(x-2)(x-3)\,dx = \int (x^3 - 6x^2 + 11x - 6)\,dx = \frac{1}{4}x^4 - 2x^3 + \frac{11}{2}x^2 - 6x.$

$\therefore \quad F(0) = 0, F(1) = -\frac{9}{4}, F(2) = -2, F(3) = -\frac{9}{4}.$

Hence required Area $= -[F(1) - F(0)] + [F(2) - F(1)] - [F(3) - F(2)] = 2\frac{3}{4}$ sq.units.

***Illustration 5 :*** Compute the area of the figure bounded by the straight lines $x = 0$, $x = 2$ and the curves $y = 2^x$, $y = 2x - x^2$.

***Solution :*** The required area $= \int_0^2 (y_1 - y_2)\,dx$

where $y_1 = 2^x$ and $y_2 = 2x - x^2 = \int_0^2 (2^x - 2x + x^2)dx$

$$= \left[\frac{2^x}{\ln 2} - x^2 + \frac{1}{3}x^3\right]_0^2 = \left(\frac{4}{\ln 2} - 4 + \frac{8}{3}\right) = \frac{1}{\ln 2} = \frac{3}{\ln 2} - \frac{4}{3} \text{ sq.units.}$$

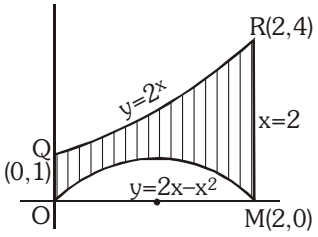



***Illustration 6 :*** Compute the area of the figure bounded by the parabolas $x = -2y^2$, $x = 1 - 3y^2$.

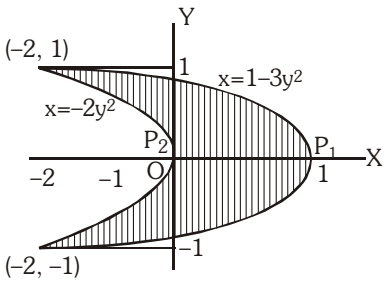
***Solution :*** Solving the equations $x = -2y^2$, $x = 1 - 3y^2$, we find that ordinates of the points of intersection of the two curves as $y_1 = -1$, $y_2 = 1$.

The points are $(-2, -1)$ and $(-2, 1)$.

The required area

$$2\int_0^1 (x_1 - x_2)\,dy = 2\int_0^1 [(1 - 3y^2) - (-2y^2)]dy$$

$$= 2\int_0^1 (1 - y^2)dy = 2\left[y - \frac{y^3}{3}\right]_0^1 = \frac{4}{3} \text{ sq.units.}$$




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**Do yourself - 2 :**

**1.** Find the area bounded by $y = \sqrt{x}$ and $y = x$.

**2.** Find the area bounded by the curves $x = y^2$ and $x = 3 - 2y^2$.

**3.** Find the area of the region bounded by the curves $x = \frac{1}{2}$, $x = 2$, $y = \log x$ and $y = 2^x$.

**4.** Find the area bounded on the right by the line $x + y = 2$, on the left by the parabola $y = x^2$ and below by the x-axis.

**5.** Find the area of the region $\{(x, y) : 0 \le y \le x^2 + 1 , 0 \le y \le x + 1 , 0 \le x \le 2\}$.

**6.** A figure is bounded by the curves $y = \left|\sqrt{2} \sin \frac{\pi x}{4}\right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x-axis straight lines must be drawn through (4, 0) so that these lines partition the figure into three parts of the same area.

**7.** The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

**8.** Find the area enclosed between $y = \sin x$ ; $y = \cos x$ and y-axis in the 1st quadrant.

## 3. CURVE TRACING :

The following procedure is to be applied in sketching the graph of a function $y = f(x)$ which in turn will be extremely useful to quickly and correctly evaluate the area under the curves.

**(a)** Symmetry : The symmetry of the curve is judged as follows :

**(i)** If all the powers of y in the equation are even then the curve is symmetrical about the axis of x.

**(ii)** If all the powers of x are even, the curve is symmetrical about the axis of y.

**(iii)** If powers of x & y both are even, the curve is symmetrical about the axis of x as well as y.

**(iv)** If the equation of the curve remains unchanged on interchanging x and y, then the curve is symmetrical about $y = x$.

**(v)** If on interchanging the signs of x & y both, the equation of the curve is unaltered then there is symmetry in opposite quadrants.

**(b)** Find dy/dx & equate it to zero to find the points on the curve where you have horizontal tangents.

**(c)** Find the points where the curve crosses the x–axis & also the y–axis.

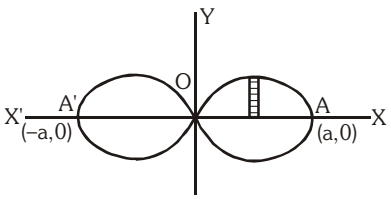
**(d)** Examine if possible the intervals when f(x) is increasing or decreasing. Examine what happens to 'y' when $x \to \infty$ or $-\infty$.

***Illustration 7 :*** Find the area of a loop as well as the whole area of the curve $a^2y^2 = x^2(a^2 - x^2)$.

***Solution :*** The curve is symmetrical about both the axes. It cuts x-axis at (0, 0), (–a, 0), (a, 0)

$$\text{Area of a loop} = 2\int_0^a y\,dx = 2\int_0^a \frac{x}{a}\sqrt{a^2 - x^2}\,dx$$

$$= -\frac{1}{a}\int_0^a \sqrt{a^2 - x^2}(-2x)dx = \frac{1}{a}\left[\frac{2}{3}(a^2 - x^2)^{3/2}\right]_0^a = \frac{2}{3}a^2$$



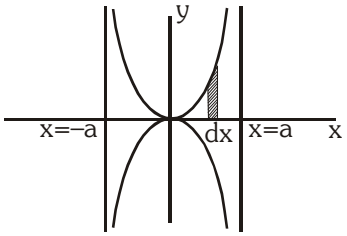

$$\text{Total area} = 2 \times \frac{2}{3}a^2 = \frac{4}{3}a^2 \text{ sq.units.}$$

***Illustration 8 :*** Find the whole area included between the curve $x^2y^2 = a^2(y^2 - x^2)$ and its asymptotes.

***Solution :*** (i) The curve is symmetric about both the axes (even powers of x & y)

(ii) Asymptotes are $x = \pm a$

$$A = 4\int_0^a y\,dx$$




$$= 4\int_0^a \frac{ax}{\sqrt{a^2 - x^2}}\,dx$$

$$= 4a\left|-\sqrt{a^2 - x^2}\right|_0^a$$

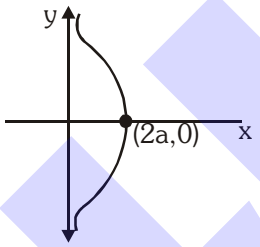
$$= 4a^2$$

***Illustration 9 :*** Find the area bounded by the curve $xy^2 = 4a^2(2a - x)$ and its asymptote.

***Solution :*** (i) The curve is symmetrical about the x-axis as it contains even powers of y.

(ii) It passes through (2a,0).

(iii) Its asymptote is $x = 0$, i.e., y-axis.




$$A = 2\int_0^{2a} y\,dx = 2\int_0^{2a} 2a\sqrt{\frac{2a - x}{x}}\,dx$$

Put $x = 2a\sin^2\theta$

$$A = 16a^2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 4\pi a^2$$

## 4. IMPORTANT POINTS :

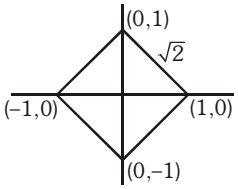
**(a)** Since area remains invariant even if the co-ordinate axes are shifted, hence shifting of origin in many cases proves to be very convenient in computing the area.

***Illustration 10 :*** Find the area enclosed by $|x - 1| + |y + 1| = 1$.

***Solution :*** Shift the origin to $(1, -1)$.

$X = x - 1$ $\qquad$ $Y = y + 1$

$|X| + |Y| = 1$



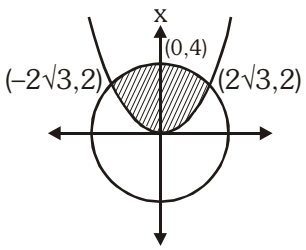

Area $= \sqrt{2} \times \sqrt{2} = 2$ sq. units

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***Illustration 11 :*** Find the area of the region common to the circle $x^2 + y^2 + 4x + 6y - 3 = 0$ and the parabola $x^2 + 4x = 6y + 14$.

***Solution :*** Circle is $x^2 + y^2 + 4x + 6y - 3 = 0$

$\Rightarrow (x + 2)^2 + (y + 3)^2 = 16$




Shifting origin to (–2,–3).

$X^2 + Y^2 = 16$

equation of parabola $\rightarrow (x + 2)^2 = 6(y + 3)$

$\Rightarrow X^2 = 6Y$

Solving circle & parabola, we get $X = \pm 2\sqrt{3}$

Hence they intersect at $\left(-2\sqrt{3}, 2\right)$ & $\left(2\sqrt{3}, 2\right)$

$$A = 2\left[\int_0^2 \sqrt{6Y}\, dY + \int_2^4 \sqrt{16 - Y^2}\, dY\right]$$

$$= 2\left[\frac{2}{3}\sqrt{6}\left[Y^{3/2}\right]_0^2 + \left[\frac{1}{2}Y\sqrt{16 - Y^2} + \frac{16}{2}\sin^{-1}\frac{Y}{4}\right]_2^4\right] = \left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3}\right) \text{sq. units}$$

**Do yourself : 3**

**1.** Find the area inside the circle $x^2 - 2x + y^2 - 4y + 1 = 0$ and outside the ellipse $x^2 - 2x + 4y^2 - 16y + 13 = 0$

**2.** Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let $D_1$ denotes the region surrounded by $C_1$, $C_2$ and the line $x = 1$ and $D_2$ denotes the region surrounded by $C_1$, $C_2$ and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.

**3.** Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to 16/3.

**4.** Compute the area of the curvilinear triangle bounded by the y-axis & the curve, $y = \tan x$ & $y = (2/3)\cos x$.

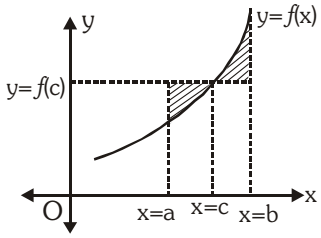
**5.** Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices (0, 0), (a, 0), $(0, a^2 + 1)$ and $(a, a^2 + 1)$.

**6.** Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$

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**(b)** If $y = f(x)$ is a monotonic function in (a, b), then the area bounded by the ordinates at x = a, x = b, $y = f(x)$ and $y = f(c)$ [where $c \in (a, b)$] is minimum when $c = \frac{a+b}{2}$.

**Proof :** Let the function $y = f(x)$ be monotonically increasing.




$$\text{Required area } A = \int_a^c [f(c) - f(x)]dx + \int_c^b [f(x) - f(c)]dx$$

For minimum area, $\frac{dA}{dc} = 0$

$$\Rightarrow \quad [f'(c).c + f(c) - f'(c)a - f(c)] + [-f(c) - f'(c).b + f'(c).c + f(c)] = 0$$

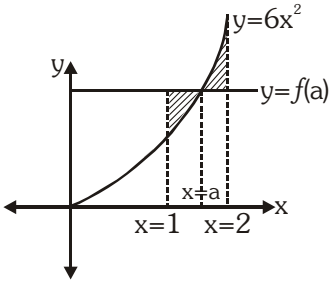
$$\Rightarrow \quad f'(c)\left\{c - \frac{a+b}{2}\right\} = 0$$

$$\Rightarrow \quad c = \frac{a+b}{2} \quad (\because f'(c) \neq 0)$$

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***Illustration 12 :*** Find the value of 'a' for which area bounded by x = 1, x=2, $y=6x^2$ and $y=f(a)$ is minimum.

***Solution :*** Let $b = f(a)$.




$$A = \int_1^a (b - 6x^2)dx + \int_a^2 (6x^2 - b)dx = \left|bx - 2x^3\right|_1^a + \left|2x^3 - bx\right|_a^2$$

$$= 8a^3 - 18a^2 + 18$$

For minimum area $\frac{dA}{da} = 0$

$$\Rightarrow \quad 24a^2 - 36a = 0 \quad \Rightarrow \quad a = 1.5$$

**Alternatively**, $y = 6x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 12x$

Hence $y = f(x)$ is monotonically increasing. Hence bounded area is minimum when

$$a = \left(\frac{1+2}{2}\right) = 1.5$$

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**Do yourself - 4 :**

**1.** Find the value of 'a' $\left(0 < a < \frac{\pi}{2}\right)$ for which the area bounded by the curve $f(x) = \sin^3 x + \sin x$, $y = f(a)$ between $x = 0$ & $x = \pi/2$ is minimum.

**2.** Find the area bounded by the curve $y = x e^{-x^2}$, the x-axis, and the line $x = c$ where $y(c)$ is maximum.

**3.** Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is, (i) 9/2 square units & (ii) minimum. Also find the minimum area.

**4.** For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ & $x = 1$ the least ?

**5.** If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is least. Find k and also the least area.

**(c)** The area bounded by a curve & an axis is equal to the area bounded by the inverse of that curve & the other axis, i.e., the area bounded by $y = f(x)$ and x-axis (say) is equal to the area bounded by $y = f^{-1}(x)$ and y-axis.

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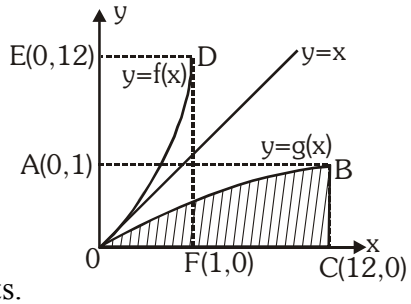
***Illustration 13 :*** If $y = g(x)$ is the inverse of a bijective mapping $f : R \to R$, $f(x) = 6x^5 + 4x^3 + 2x$, find the area bounded by g(x), the x-axis and the ordinate at $x = 12$.

***Solution :*** $f(x) = 12$

$\Rightarrow \quad 6x^5 + 4x^3 + 2x = 12 \quad \Rightarrow \quad x = 1$

$$\int_0^{12} g(x)dx = \text{area of rectangle OEDF} - \int_0^1 f(x)dx$$

$$= 1 \times 12 - \int_0^1 (6x^5 + 4x^3 + 2x)dx = 12 - 3 = 9 \text{ sq. units.}$$




**Do yourself - 5 :**

**1.** Find the area bounded by the inverse of bijective function $f(x) = 4x^3 + 6x$, the x-axis and the ordinates $x = 0$ & $x = 44$.

**2.** Find area bounded by $y = f(x)$ & $y = f^{-1}(x)$, $x = 1$ & $x = e$ where $f(x) = e^x$.

**3.** Let $f(x) = x^3 + 3x + 2$ and g(x) is the inverse of it. Find the area bounded by g(x), the x-axis and the ordinate at $x = -2$ and $x = 6$.

**4.** Find area bounded by $y = f^{-1}(x)$, $x = 10$, $x = 4$ and x-axis given that area bounded by $y = f(x)$, $x = 2$, $x = 6$ and x-axis is 30 sq. units, where $f(2) = 4$ and $f(6) = 10$. (given f(x) is an invertible function)

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## 5. USEFUL RESULTS :

**(a)** Whole area of the ellipse, $x^2/a^2 + y^2/b^2 = 1$ is $\pi$ ab sq.units.

**(b)** Area enclosed between the parabolas $y^2 = 4ax$ & $x^2 = 4by$ is 16ab/3 sq.units.

**(c)** Area included between the parabola $y^2 = 4ax$ & the line $y = mx$ is $8a^2/3m^3$ sq.units.

**(d)** The area of the region bounded by one arch of sin ax (or cos ax) and x-axis is 2/a sq.units.

**(e)** Average value of a function $y = f(x)$ over an interval $a \le x \le b$ is defined as : $y(av) = \frac{1}{b-a}\int_a^b f(x)dx$.

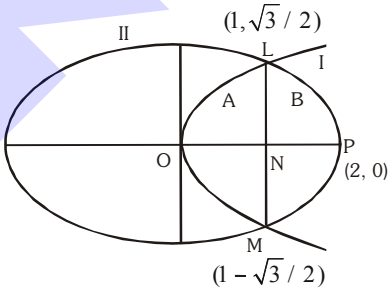
***Miscellaneous Illustration :***

***Illustration 14 :*** Find the smaller of the areas bounded by the parabola $4y^2 - 3x - 8y + 7 = 0$ and the ellipse $x^2 + 4y^2 - 2x - 8y + 1 = 0$.

***Solution :*** $C_1$ is $4(y^2 - 2y) = 3x - 7$

or $4(y-1)^2 = 3x - 3 = 3(x-1)$ .....(i)

Above is parabola with vertex at (1, 1)




$C_2$ is $(x^2 - 2x) + 4(y^2 - 2y) = -1$

or $(x-1)^2 + 4(y-1)^2 = -1 + 1 + 4$

or $\frac{(x-1)^2}{2^2} + \frac{(y-1)^2}{1^2} = 1$ .....(ii)

Above represents an ellipse with centre at (1, 1). Shift the origin to (1, 1) and this will not affect the magnitude of required area but will make the calculation simpler.

Thus the two curves are

$4Y^2 = 3X$ and $\frac{X^2}{2^2} + \frac{Y^2}{1} = 1$ They meet at $\left(1, \pm\frac{\sqrt{3}}{2}\right)$

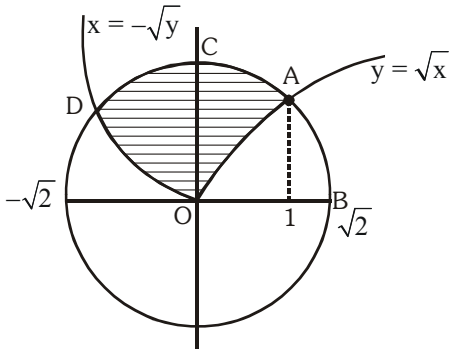
Required area $= 2(A + B) = 2\left[\int Y_1 dX + \int Y_2 dX\right]$

$$= 2\left[\frac{\sqrt{3}}{2}\int_0^1 \sqrt{X}dX + \int_1^2 \frac{\sqrt{4 - X^2}}{2}dX\right] = \left[\frac{\sqrt{3}}{6} + \frac{2\pi}{3}\right] \text{ sq.units.}$$

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***Illustration 15 :*** Find the area bounded by the regions $y^2 \geq x$, $x > -\sqrt{y}$ & curve $x^2 + y^2 \leq 2$.

***Solution :*** Common region is given by the diagram




If area of region OAB = $\lambda$

then area of OCD = $\lambda$

Because $y = \sqrt{x}$ & $x = -\sqrt{y}$

will bound same area with x & y axes respectively.

$y = \sqrt{x} \Rightarrow y^2 = x$

$x = -\sqrt{y} \Rightarrow x^2 = y$ and hence both the curves are symmetric with respect to the line $y = x$

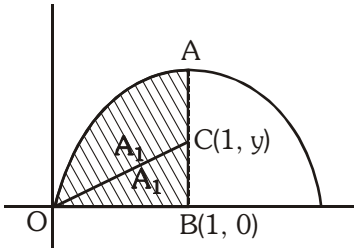
Area of first quadrant OBC $= \frac{\pi r^2}{4} = \frac{\pi}{2}$ $(\because\ r = \sqrt{2})$

Area of region OCA $= \frac{\pi}{2} - \lambda$

Area of shaded region $= (\frac{\pi}{2} - \lambda) + \lambda = \frac{\pi}{2}$ sq.units.

***Illustration 16 :*** Find the equation of line passing through the origin & dividing the curvilinear triangle with vertex at the origin, bounded by the curves $y = 2x - x^2$, $y = 0$ & $x = 1$ in two parts of equal areas.

***Solution :*** Area of region OBA $= \int_0^1 (2x - x^2)dx$




$$= \left[x^2 - \frac{x^3}{3}\right]_0^1 = \frac{2}{3}$$

$\frac{2}{3} = A_1 + A_1 \Rightarrow A_1 = \frac{1}{3}$

Let pt. C has coordinates (1, y)

Area of $\Delta OCB = \frac{1}{2} \times 1 \times y = \frac{1}{3}$

$y = \frac{2}{3}$

C has coordinates $\left(1, \frac{2}{3}\right)$

Line OC has slope $m = \frac{\frac{2}{3} - 0}{1 - 0} = \frac{2}{3}$

Equation of line OC is $y = mx \Rightarrow y = \frac{2}{3}x$.

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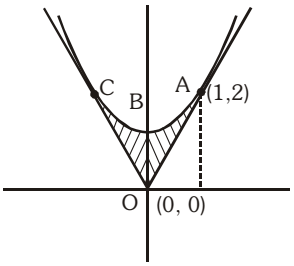
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***Illustration 17 :*** The area bounded by $y = x^2 + 1$ and the tangents to it drawn from the origin is :-

(A) 8/3 sq. units (B) 1/3 sq. units (C) 2/3 sq. units (D) none of these

***Solution :*** The parabola is even function & let the equation of tangent is $y = mx$

Now we calculate the point of intersection of parabola & tangent




$mx = x^2 + 1$

$x^2 - mx + 1 = 0 \Rightarrow D = 0$

$\Rightarrow m^2 - 4 = 0 \Rightarrow m = \pm 2$

Two tangents are possible $y = 2x$ & $y = -2x$

Intersection of $y = x^2 + 1$ & $y = 2x$ is $x = 1$ & $y = 2$

$$\text{Area of shaded region OAB} = \int_0^1 (y_2 - y_1)\,dx = \int_0^1 ((x^2+1) - 2x)\,dx = \frac{1}{3} \text{ sq. units}$$

$$\text{Area of total shaded region} = 2\left(\frac{1}{3}\right) = \frac{2}{3} \text{ sq. units}$$

***Illustration 18 :*** Find the area bounded by y-axis and the curve given by $x = a\sin t$, $y = a\cos t$ for $0 \le t \le \pi$.

***Solution :***

$$\text{Area} = \int_0^{\pi} y\frac{dx}{dt}.dt = \int_0^{\pi} a\cos t(a\cos t)dt = \frac{a^2}{2}\int_0^{\pi}(1+\cos 2t)dt = \frac{a^2}{2}\left|t + \frac{\sin 2t}{2}\right|_0^{\pi} = \frac{a^2}{2}|\pi| = \frac{\pi a^2}{2}$$

Alternatively,

$$\text{Area} = \int_0^{\pi} x\frac{dy}{dt}.dt = \left|\int_0^{\pi} a\sin t(-a\sin t)dt\right| = \frac{a^2}{2}\left|\int_0^{\pi}(\cos 2t - 1)dt\right| = \frac{a^2}{2}\left|-\frac{\sin 2t}{2} - t\right|_0^{\pi} = \frac{\pi a^2}{2}$$

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